Assignment 1

Aayush M Anand
21d170001

1.Statistical Measures

count : 11712.00

mean : 3889.10

median : 3240.04

mode : 10000.00

min : 0.00

max : 10000.00

variance : 6383347.39

std_dev : 2526.53

percentile_25 : 2349.40

percentile_50 : 3240.04

percentile_75 : 4200.56

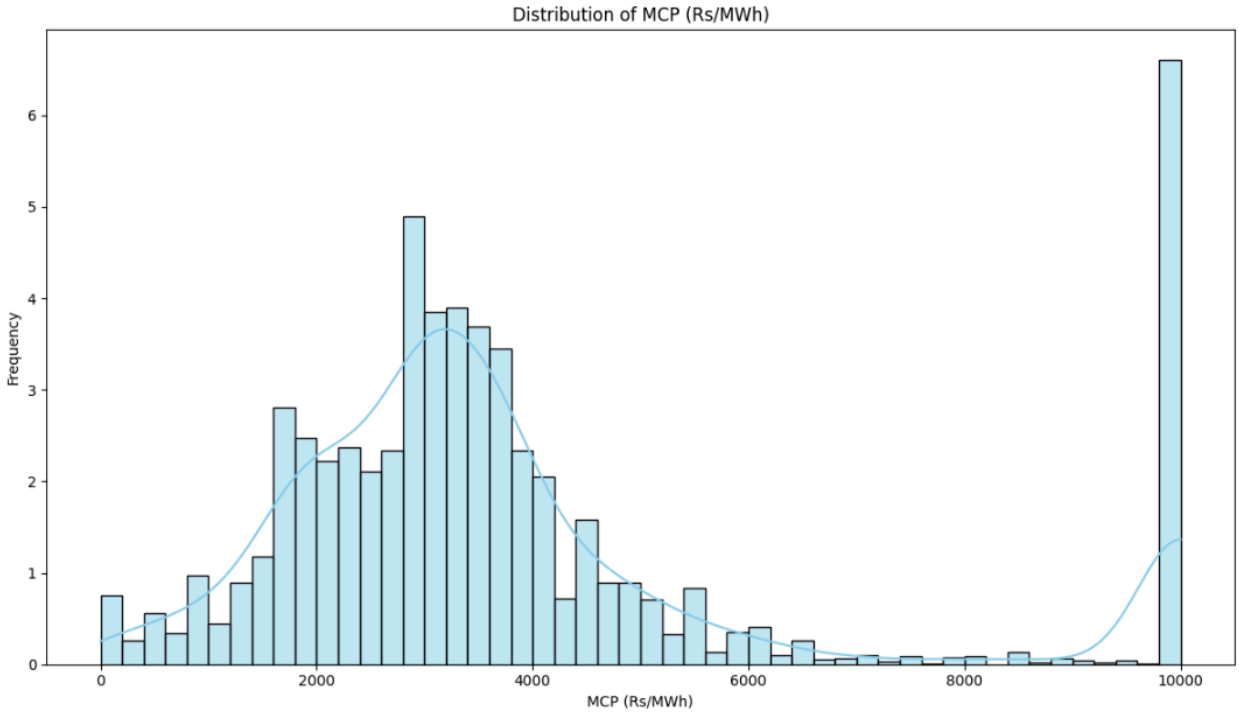
2.Per day statistical measures

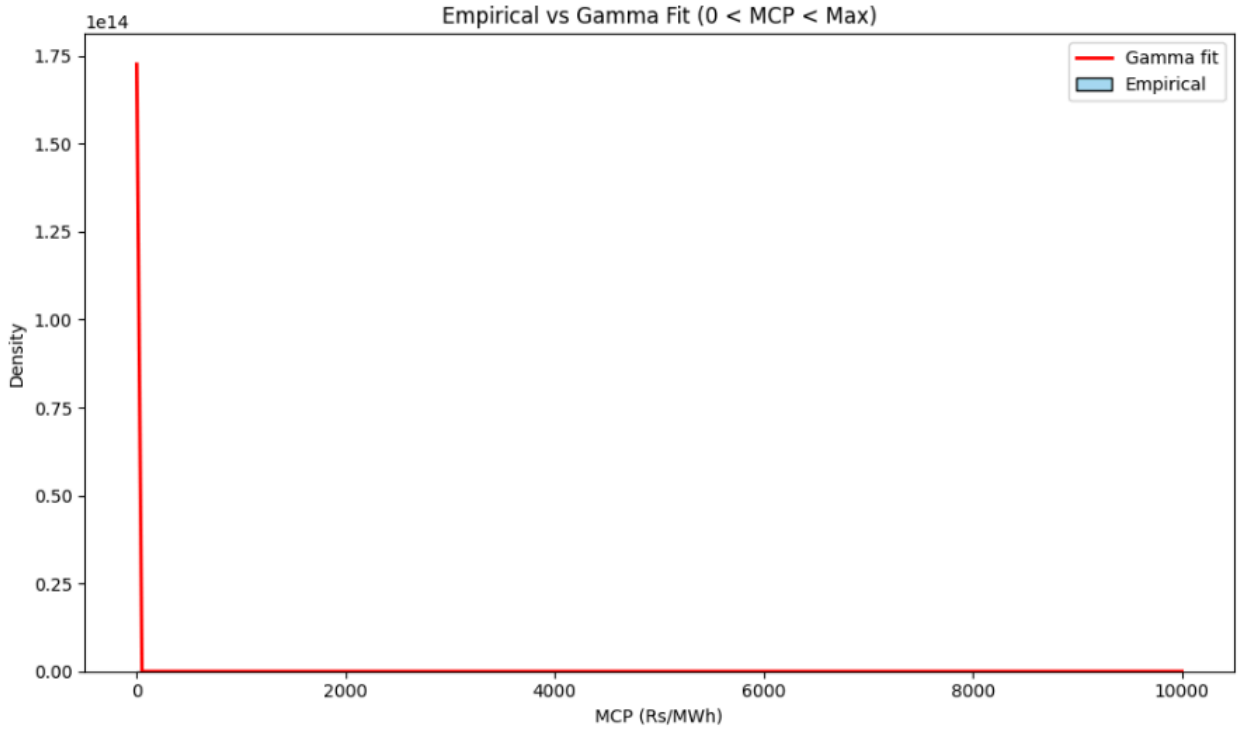
date	count	mean	median	min	max	stddev	perc25	perc75
2025-04-01	96	3780.805	3451.115	1583.87	10000	1806.078	2808.253	4219.438
2025-04-02	96	4862.729	3990.195	1999.45	10000	2302.366	3195.15	5834.793
2025-04-03	96	4570.913	3751.435	1499.91	10000	2333.178	3147.603	4999.37
2025-04-04	96	3864.994	3510.4	0	10000	2222.372	2466.445	4520.14
2025-04-05	96	4100.352	3973.885	1550.52	10000	1999.232	2717.278	4579.74

Insights from the above data

- Many days show price spikes reaching the cap of 10 Rs/MWh, often during morning and evening hours reflecting demand surges.
- Low price periods commonly occur overnight or early morning when demand is low.
- Some days feature moderate, stable prices mostly between 2 and 4 Rs/MWh.
- Larger spreads between minimum and maximum prices indicate higher daily volatility consistent with market stress

3. Probability distribution model of MCP data





Gamma distribution fit parameters (excluding spikes at min and max):
Shape: 0.02, Loc: 0.34, Scale: 3.35

- The MCP distribution is strongly right-skewed and bimodal.
- Primary cluster of prices lies between roughly 2-4 Rs/MWh.
- A discrete spike at the maximum price of 10 Rs/MWh shows a price cap effect.
- Another smaller cluster near 0 Rs/MWh exists, reflecting very low-price periods.
- Fitting a Gamma distribution to the MCP values excluding zero and the capped max yields parameters:
    - Shape - 15.63
    - Location - -2.34
    - Scale - 0.35
- This confirms the main body of the data follows a right-skewed continuous distribution, with discrete probability masses at 0 and the cap.
- The mean being greater than the median and mode coinciding with the cap price reinforces the presence of volatility and skew